

Name: _____ **Section:** _____ **Date:** _____

	Part I	Part II	Part III	Part IV	Part V	Total
Points	30	20	20	10	20	100
Score						

_____ 6. Which is the correct order of the three PCM steps?

- A. Quantization → sampling → encoding
- B. Sampling → quantization → binary encoding
- C. Encoding → sampling → quantization
- D. Sampling → encoding → quantization

_____ 7. Binary ASK with $d = 1$ must fit in an available bandwidth of 40 kHz. What is the maximum bit rate?

- A. 20 kbps
- B. 10 kbps
- C. 40 kbps
- D. 80 kbps

_____ 8. The power of a signal is 2 mW and the power of the noise is 2 μ W. What is SNR_{dB} ?

- A. 3 dB
- B. 20 dB
- C. 25 dB
- D. 30 dB

_____ 9. A composite signal is made of sine waves with frequencies f , $3f$ and $9f$, where $f = 200$ Hz. What is its bandwidth?

- A. 1,600 Hz
- B. 1,800 Hz
- C. 200 Hz
- D. 600 Hz

_____ 10. In asynchronous transmission, each byte is framed by:

- A. a start bit 1 and a stop bit 0
- B. a start bit 0 and one or more stop bits 1
- C. a flag byte at each end
- D. no extra bits at all

_____ 11. In an mBnL code, $2^m \leq L^n$ must hold. Which of these codes is **not** valid?

- A. 2B1Q
- B. 8B6T
- C. 3B1Q
- D. 4B3T

_____ 12. HDB3 substitutes four zeros with **000V** when the number of nonzero pulses since the last substitution is odd. What does it use when that number is **even**?

- A. 000V
- B. 000VB0VB
- C. B00V
- D. 0000 (no substitution)

_____ 13. What is **baseline wandering**?

- A. A slow change in the sender's clock frequency
- B. A drift of the receiver's running-average baseline, caused by a long string of 0s or 1s
- C. The delay between the first and the last bit of a block
- D. Noise that the channel adds to the signal

_____ 14. Which OSI layer frames the data and moves frames error-free between two end nodes over the physical layer?

- A. Physical
- B. Data link
- C. Network
- D. Transport

_____ 15. A system uses NRZ-I to send data at 2 Mbps. What is the minimum bandwidth (average case)?

- A. 1 MHz
- B. 500 kHz
- C. 2 MHz
- D. 4 MHz

_____ 16. According to the slides, a LAN typically connects computers within:

- A. 50 m
- B. 500 m
- C. 5 km
- D. 50 km

_____ 17. Which IEEE standard defines wireless media access control and physical layer specifications (the Wi-Fi family)?

- A. 802.3
- B. 802.5
- C. 802.11
- D. 802.16

_____ 18. Data at 8 Mbps is block-coded with 4B/5B before line coding. What is the bit rate on the line after block coding?

- A. 6.4 Mbps
- B. 8 Mbps
- C. 16 Mbps
- D. 10 Mbps

_____ 19. A periodic signal has a period of 4 ms. What is its frequency?

- A. 4 Hz
- B. 4 kHz
- C. 25 Hz
- D. 250 Hz

_____ 20. Which statement about signals is correct?

- A. Only analog signals can be periodic
- B. A nonperiodic signal repeats its pattern in every period
- C. Both analog and digital signals can be periodic or nonperiodic
- D. Digital signals can never be periodic

_____ 21. Why is PSK more robust than ASK?

- A. PSK uses a higher carrier frequency
- B. PSK needs no carrier
- C. PSK has a lower baud rate
- D. Noise mostly alters amplitude, and PSK keeps the amplitude constant

_____ 22. Using the slides' QPSK mapping (first bit → I carrier), which dibit is sent at phase **135°**?

- A. 00
- B. 11
- C. 01
- D. 10

_____ 23. Which description of **AMI** matches the slide notes?

- A. No DC component or baseline wandering, but no self-synchronization during a long run of 0s
- B. A DC component, but good self-synchronization
- C. Self-synchronization and error detection, but a DC component
- D. No DC component and a transition in the middle of every bit

_____ 24. A signal travels through a cable and its power is reduced to one-half. The change is about:

- A. -3 dB
- B. -6 dB
- C. +3 dB
- D. -0.5 dB

_____ 25. A low-pass signal contains frequencies from 0 to 5 kHz. What is the minimum sampling rate?

- A. 5000 samples/s
- B. 2500 samples/s
- C. 20,000 samples/s
- D. 10,000 samples/s

_____ 26. A signal is quantized into $L = 32$ levels and sampled at 8000 samples/s. What is the bit rate?

- A. 40 kbps
- B. 256 kbps
- C. 32 kbps
- D. 64 kbps

_____ 27. Which line code has a level transition in the middle of **every** bit and uses only **two** voltage levels?

- A. RZ
- B. Manchester
- C. AMI
- D. NRZ-I

_____ 28. An analog signal carries 3 bits per signal element at 2000 baud. What is the bit rate?

- A. 667 bps
- B. 2000 bps
- C. 6000 bps
- D. 8000 bps

_____ 29. In the slides' TCP/IP model, which layer corresponds to OSI layers 1 and 2?

- A. Internet
- B. Transport
- C. Application
- D. Network access

_____ 30. Compared with NRZ, the **minimum bandwidth** of Manchester coding is:

- A. Half as much
- B. The same
- C. One and a half times as much
- D. Twice as much

Part II — Identification

20 questions × 1 pt = 20 pts

Write the term on the answer line.

1. The third step of block coding, after division and substitution, in which the n -bit groups are joined back into one stream.

Answer: _____

2. The scrambling technique that replaces eight consecutive zeros with 000VB0VB.

Answer: _____

3. The data link layer service of partitioning data into frames with recognized frame boundaries.

Answer: _____

4. The block code whose encoder splits each 8-bit block between a 5B/6B encoder and a 3B/4B encoder and passes the result through a disparity controller.

Answer: _____

5. The sampling method that holds each sample at one constant amplitude for the pulse duration.

Answer: _____

6. The polar scheme that uses three voltage values (+, 0, -) and has a transition in the middle of every symbol, from high or low to zero.

Answer: _____

7. The form of binary ASK in which a 1 sends the carrier and a 0 sends no signal at all.

Answer: _____

8. A measure of how fast we can actually send data through a network (always less than the link's bandwidth).

Answer: _____

9. The loss of some of a signal's energy as it travels through a medium, in overcoming the resistance of the medium.
Answer: _____
10. The kind of noise in which signals jump from one wire to another, normally between adjacent wires.
Answer: _____
11. A change in the form or shape of a composite signal because its frequency components propagate at different speeds.
Answer: _____
12. The digital-to-analog scheme that changes both the amplitude and the phase of the carrier.
Answer: _____
13. The diagram that shows each signal element as a point whose distance from the origin is its amplitude and whose angle is its phase.
Answer: _____
14. The technique that compresses sample values at the sender and expands them at the receiver to keep the SNQR more even across amplitudes.
Answer: _____
15. The kind of noise caused by the random motion of electrons in a wire.
Answer: _____
16. In the 4B/5B table, the control sequence **11111** is named:
Answer: _____
17. The coverage unit a wireless network uses to interconnect mobile devices.
Answer: _____
18. The OSI layer that translates frames into electrical, optical or electromagnetic signals representing 0s and 1s.
Answer: _____
19. The digitization scheme that sends a single bit per sample: 1 if the signal rose above the staircase, 0 if it did not.
Answer: _____
20. The term for unused fiber-optic cable.
Answer: _____

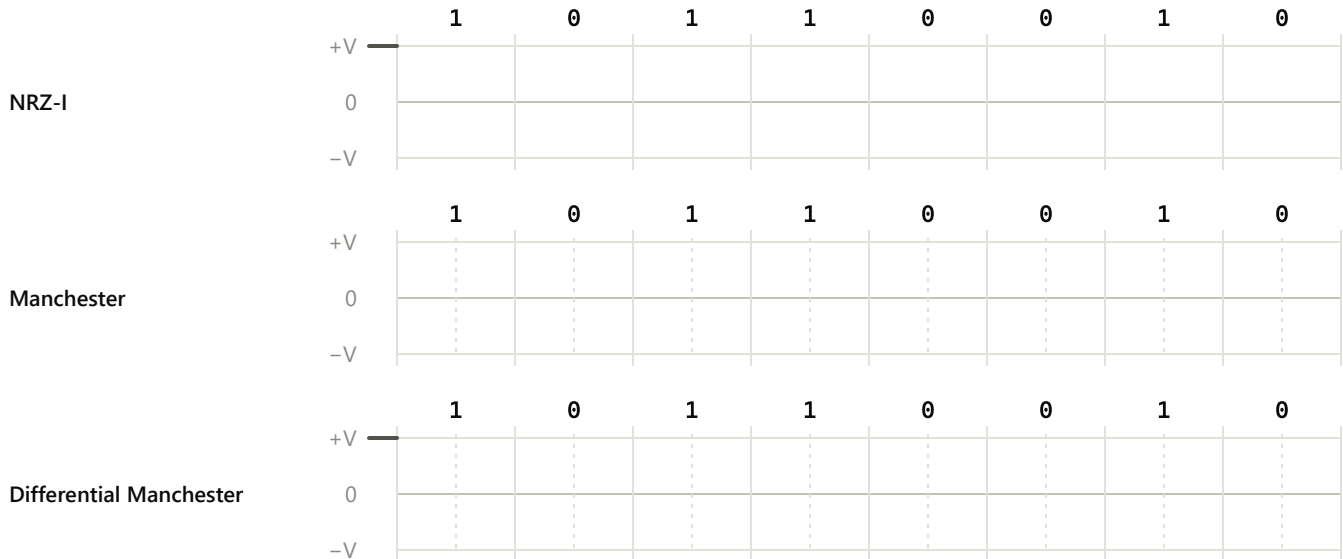
Part III — Draw: line encodings and waves

4 problems × 5 pts = 20 pts

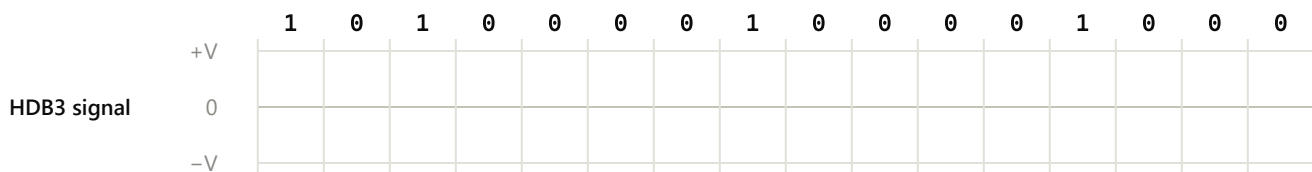
Show a complete solution; give final answers with units.

1. Draw the bit stream **10110010** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions: [5 pts]

- NRZ-I: a 1 inverts the level at the start of the bit, a 0 does not; assume the level before the first bit is +V.
- Manchester: 0 = high→low, 1 = low→high (a transition in the middle of every bit).
- Differential Manchester: always a transition in the middle of the bit; a 0 also has a transition at the start of the bit, a 1 does not; assume the level before the first bit is +V.



2. Apply **HDB3** scrambling (on top of AMI) to the data stream **1010 0001 0000 1000** and draw the line signal. [5 pts]
The last nonzero pulse sent before this stream was negative (−V). At the start of the stream the number of nonzero pulses since the last substitution is even.



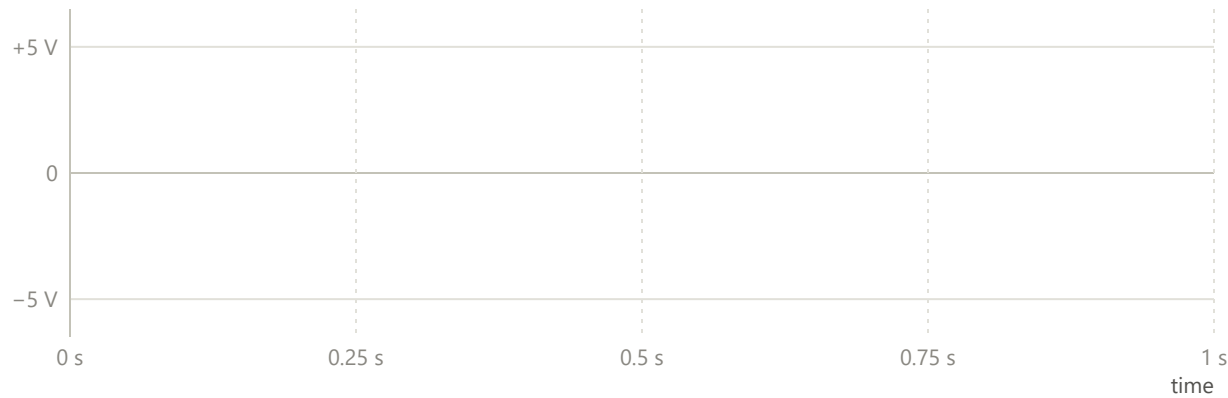
3. Draw the **binary FSK** signal for the bits **1011**: a 1 uses the higher frequency $f_1 = f_c + \Delta f$, a 0 the lower frequency $f_2 = f_c - \Delta f$ (draw about 4 cycles per bit for f_1 and 2 for f_2). [5 pts]

Binary FSK for 1011



4. Draw one second of a sine wave with peak amplitude **5 V**, frequency **2 Hz** and phase **90°**. [5 pts]

Sine wave, 0 to 1 s



Part IV — Problem solving (no calculator)

2 problems × 5 pts = 10 pts

Show a complete solution; give final answers with units.

1. We have a channel with a **2 MHz** bandwidth. The SNR for this channel is **255**. Find the upper limit of the bit rate. For better performance we then use a lower bit rate of **12 Mbps**: how many signal levels are needed? [5 pts]

WORK

Upper limit C = _____ bps

Signal levels L = _____

2. A low-pass analog signal with frequencies up to **15 kHz** is digitized with PCM using **64 quantization levels**, sampled at the Nyquist rate. Find the sampling rate, the bits per sample, the bit rate, and the minimum bandwidth of the digitized signal (NRZ-type line code). [5 pts]

WORK

Sampling rate f_s = _____ sps

Bits per sample n_b = _____

Bit rate N = _____ bps

Minimum bandwidth = _____ Hz

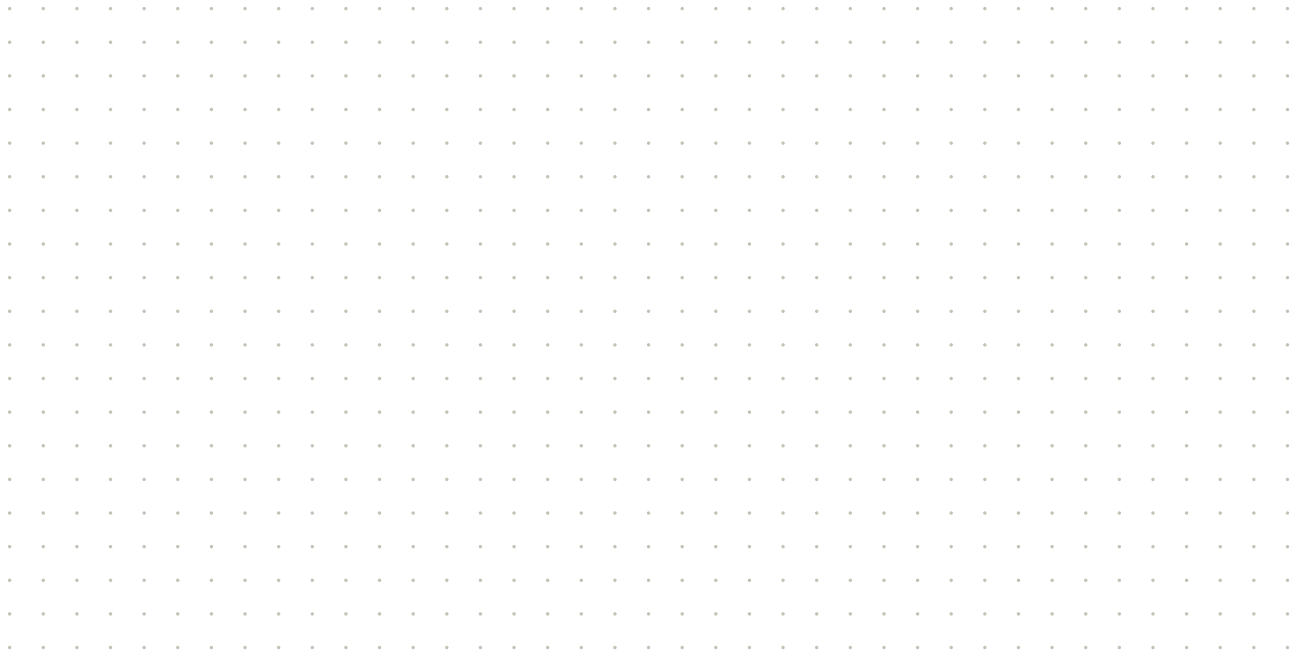
Part V — Essay (draw and/or explain)

4 questions × 5 pts = 20 pts

Answer in the space provided — draw, explain, or both.

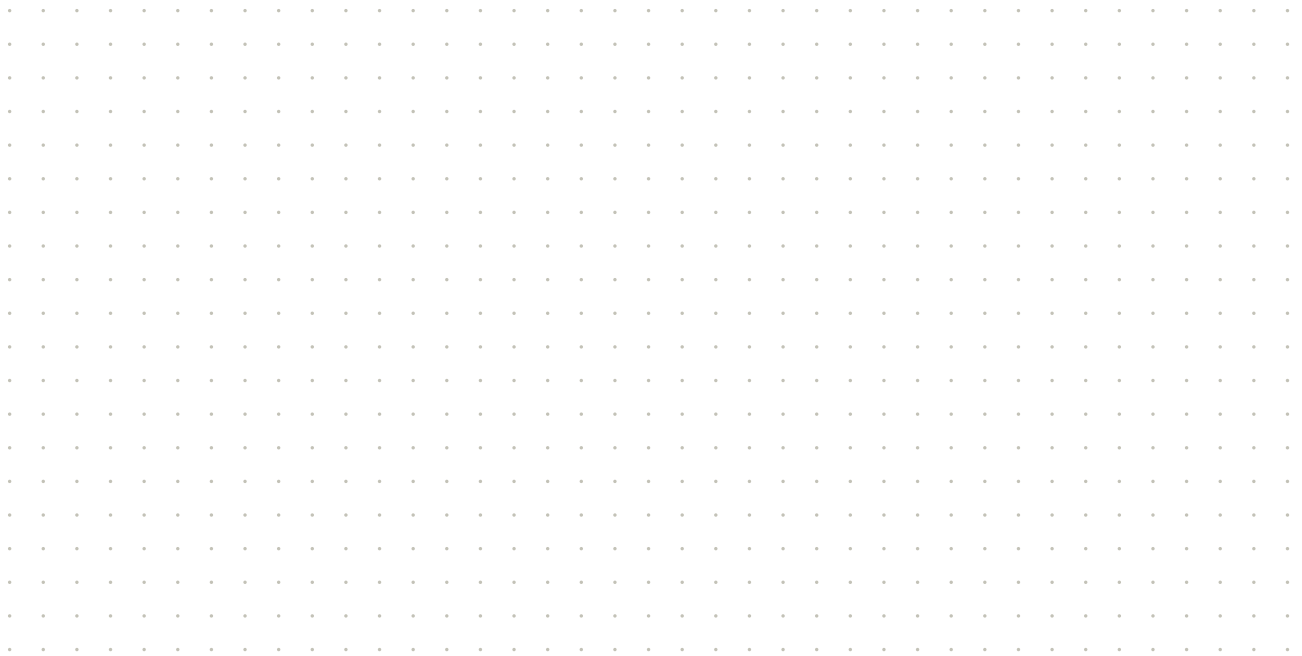
1. Draw the **OSI** model and the slides' **TCP/IP** model side by side, showing which OSI layers each TCP/IP layer covers. Then explain the job of the two lowest OSI layers. [5 pts]

DRAW AND/OR EXPLAIN



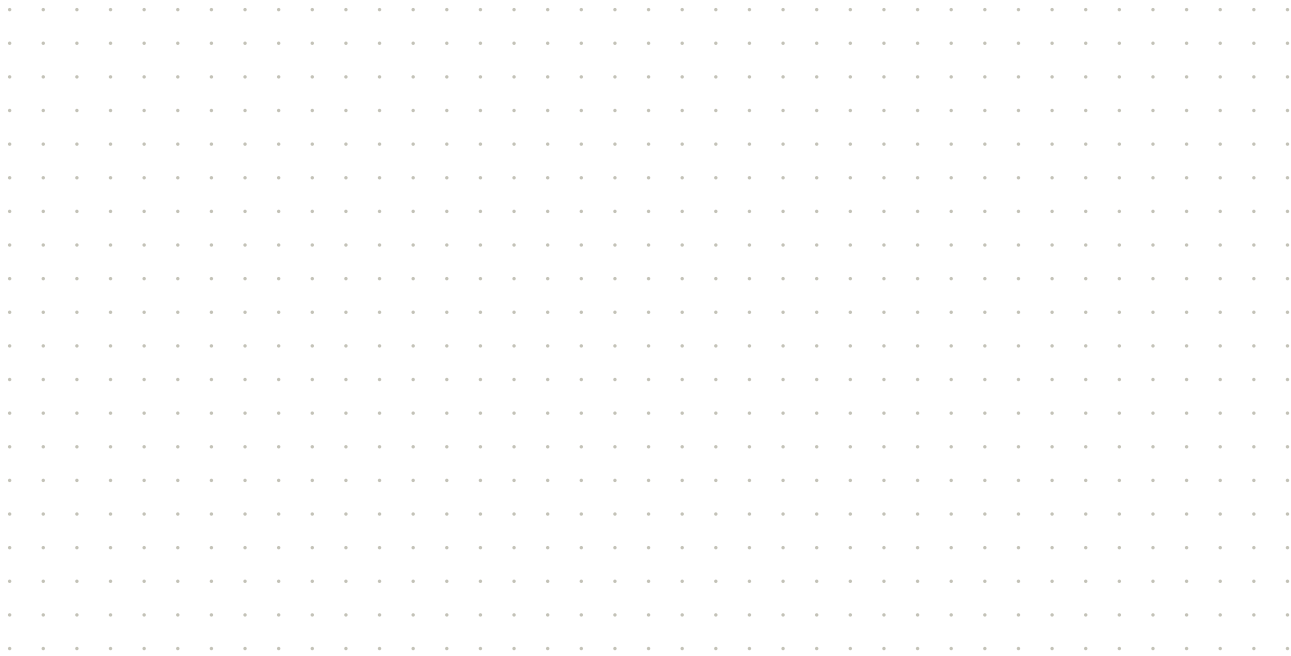
2. Explain how **PCM** turns an analog signal into a digital one. Draw a short analog signal and show what each of the three steps does to it, then explain why the sampling rate must be at least twice the highest frequency. You may draw, explain, or both. [5 pts]

DRAW AND/OR EXPLAIN



3. Compare **NRZ-L**, **NRZ-I**, **Manchester** and **AMI** for three properties: **baseline wandering**, **DC component** and **self-synchronization**. For each code, say whether it has the problem and explain *why*. You may draw a short bit stream (for example a 1 followed by a long run of 0s) in each code to support your answer. [5 pts]

DRAW AND/OR EXPLAIN



4. Compare the **Nyquist** and **Shannon** formulas for the data-rate limit of a channel. For each, state the formula, when it applies and what it depends on. Then explain how the two are used together to choose a bit rate and a number of signal levels, using a channel with a 1 MHz bandwidth and an SNR of 63 as your example. [5 pts]

DRAW AND/OR EXPLAIN

